

Makendy Midouin

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EDUCATION

- New York University**, Master of Science in Computer Science | GPA: 3.45 May 2025
- **NYU Entrepreneurial Institute**: Founded COLEAGUE; validated product-market fit via 30+ user interviews and executed a strategic product pivot based on identified market gaps
 - **Relevant Coursework**: Algorithms, Database Systems, Big Data (Spark/Hadoop), AI, Software Engineering
- St. Thomas Aquinas College**, Bachelor of Science in Computer Science | GPA: 3.68 May 2022

SKILLS

Languages: JavaScript/TypeScript, Java, Python, SQL
Frameworks: React, Next.js, Node.js, Express, Django, Flask
Cloud & Data: AWS (S3, EC2, Lambda, DynamoDB, API Gateway), MySQL, PostgreSQL, MongoDB
Tools & Practices: Docker, Git, CI/CD (Travis CI, GitHub Actions), REST/GraphQL APIs, Unit & Integration Testing (Jest, PyTest)

PROFESSIONAL EXPERIENCE

- Adjunct Faculty: Software Engineering & Cloud Computing** | *St. Thomas Aquinas College* Aug 2024 – Dec 2025
- Designed curriculum and instructed 40+ students in Cloud Architecture (AWS), CI/CD pipelines, and Full-Stack development
 - Mentored students through technical project lifecycles, emphasizing production-ready code standards and Agile methodologies
- Software Engineer** | *Roadcall.co* June 2023 – Dec 2023
- Built Django-based carrier portal with React/Redux frontend, implementing badge systems, pagination, and load forms handling 5000+ daily submissions
 - Resolved 90% (135/150) of reported issues by analyzing server logs and reproducing API errors to identify root causes
 - Implemented comprehensive unit and integration testing that resolved 10+ critical defects in legacy code, eliminating data corruption issues
 - Designed and implemented a notes feature using React and API endpoints, supporting communication across 20 operations teams
 - Presented technical solutions to non-technical stakeholders and coordinated sprint delivery in Agile environment
- Software Engineer** | *Treehouse Strategy* May 2021 – Sept 2021
- Optimized backend API call structure and caching, reducing dashboard load times by 40% for agency client portals
 - Built a real-time dashboard page for agency partnerships using React, increasing client engagement and interaction
 - Integrated REST APIs with responsive data tables, improving visualization and performance for client portals
 - Implemented reusable React components, reducing bugs and improving application maintainability across the platform
- Computer Science Tutor** | *Varsity Tutors & St. Thomas Aquinas College* Sept 2020 – May 2022
- Tutored 100+ students in Java, Data Structures, Algorithms, and Object-Oriented Programming, achieving 90% pass rate
 - Maintained 4.9/5.0 student satisfaction rating with consistent positive feedback on teaching clarity
- XLAB Researcher** | *St. Thomas Aquinas College* Sept 2019 – May 2022
- Developed a VR app using Unity and C# to study unconscious bias, implementing modular dialogue systems and Motion Capture integration for immersive interactions
 - Led a team of 5 researchers using Agile/Scrum to coordinate development sprints and meet national conference deadlines
 - Selected as Keynote Speaker for an institutional fundraiser, delivering a technical presentation that helped secure \$125,000 in STEM scholarships for underrepresented students

PROJECTS

- SmartFRQ – Full Stack Developer** [Link](#)
- Built an AI-powered IDE to help AP CS A students tackle exam-style Free-Response Questions through Socratic debugging, serving 10+ students with 100+ completed practice sessions; developed with Next.js, TypeScript, Supabase, and GPT-4o
 - Architected an adaptive learning engine with LLM verification layer to identify coding misconceptions and generate personalized quizzes
- NYU Interview Prep – Full Stack Developer** [GitHub](#)
- Led a team of 5 building an interview platform with React, Django, and PostgreSQL for recording interviews and peer feedback
 - Developed a video upload system with AWS S3 integration, engineered an in-browser code editor for technical questions, and implemented a CI/CD pipeline using Coveralls.io and Travis CI for automated testing and deployment
- Stack Overflow Post Big Data - Full Stack Developer** [GitHub](#)
- Built analytics dashboard using Next.js, Python, and MongoDB to process 100GB StackOverflow dataset with tag popularity tracking, hourly activity analysis, and TF-IDF similarity matching
 - Optimized MongoDB performance by limiting queries to 500,000 documents and caching top 1,000 tags in AWS DynamoDB for faster retrieval

LEADERSHIP EXPERIENCE

Brains & Motion Education – Site Director & Lead Coding Instructor Summer 2024 & 2025